

9708/feb march/ 2023

1-D:

A rational consumer is characterized by certain assumptions that guide their decision-making process.

Let's examine each assumption individually: → Access to complete information: A rational consumer is assumed to have access to complete information regarding the available choices, their prices, and the consequences of those choices. They are well-informed about the features, benefits, and potential drawbacks of the products or services they are considering. This assumption allows the consumer to make informed decisions based on their preferences and goals.

→ Absence of addictive behavior: Rational consumer behavior assumes that individuals do not exhibit addictive behaviors that impair their ability to make reasoned choices. This means that the consumer can evaluate alternatives objectively, without being unduly influenced by impulsive or addictive tendencies. It assumes that consumers can exercise self-control and make choices based on their long-term best interests rather than succumbing to short-term gratification.

→ Choice independent of other consumers: Rational consumer behavior assumes that an individual's decision-making process is independent of the choices made by other consumers. This means that the consumer's preferences, tastes, and decisions are not influenced by the actions or choices of others. They make choices based on their own utility function and personal preferences, without being swayed by social pressure or the behavior of others

2-B:

If a consumer wants to maximize total utility, he must be operating at the consumer equilibrium, whereby equi-marginal principle is established

It is when :

$$\rightarrow MU_S/P_S = MU_T/P_T$$

The price of good S is \$6 and the price of good T is \$8. A consumer's marginal utility for each good is shown.

MU_S/P_S	units	marginal utility of good S	marginal utility of good T	MU_T/P_T
6	1	36	32	4
4	2	24	22	2.75
3	3	18	20	2.5
1.5	4	9	18	2.25
0	5	0	9	1.125

If this consumer wants to maximise total utility, which combination of goods would they consume?

	good S	good T
A	1	1
B	2	1
C	3	4
D	5	5

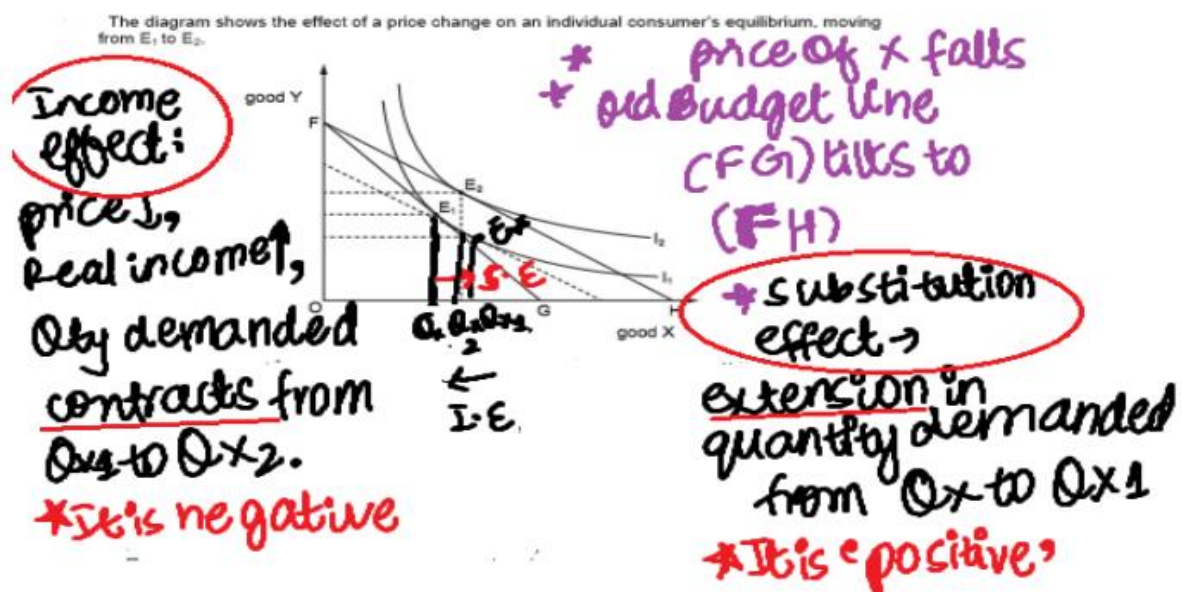
at qty '2'

$$\frac{MU_S}{P_S} = \frac{MU_T}{P_T}$$

↳ 4 = 4

at qty '1'

3-B:



4-D:

Reducing taxes on retained profits encourages firms to re-invest more of their retained profits in research and development, adoption of new technologies, and innovation, leading to dynamic efficiency. This results in Long run Average cost being decreased at a particular output level over a period of time. **This explains why key D will be correct.**

A) Reducing entry barriers increases competition, which is most likely to encourage **productive efficiency**.

B) Legally binding firms to produce at the point where price equals marginal cost will result in **allocative efficiency to be achieved**.

C) Raising taxes on profits may discourage firms from investing in innovation and technology, leading to **reduced dynamic efficiency**.

5-D: **Sales maximization is achieved when the highest number of units are sold. This occurs when the firm charges the lowest price it possibly can. It charges the lowest possible price, when it makes "normal profits", that are made when Average revenue=Average cost.**

Note:- Normal profits, also known as zero economic profits, refer to the level of profit earned by a firm that is just sufficient to cover all of its costs, including both explicit costs (such as wages, rent, and raw materials) and implicit costs (such as the opportunity cost of the owner's time and capital). In other words, normal profits represent the minimum level of profit required to keep a firm operating in the long run. Normal profits are considered the baseline level of profit that a firm needs to earn to remain in business.

A – Here, the equilibrium is achieved when $MC = MR$, that is when, a profit-maximizing level of output is produced. Here, the least qty of "A" will be sold.

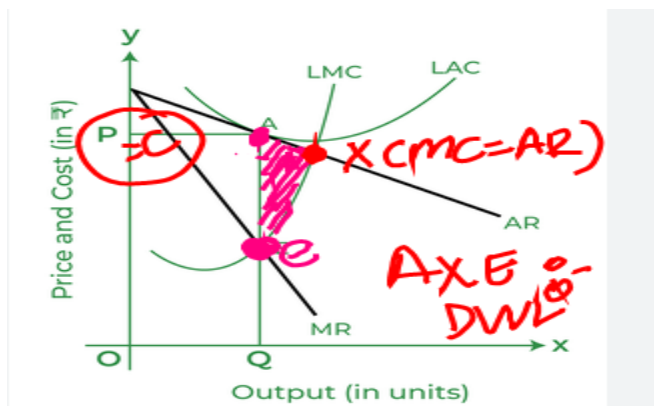
B – Here, the equilibrium is achieved when Marginal Revenue is zero, or in other words, sales revenue is maximized. Here, a relatively higher qty of "B" will be sold.

C – Here, the equilibrium is achieved when $MC=AC$, or in other words, when Average cost is at its minimum, which means that “productive efficiency is achieved”. Here, also the qty of “C” is sold, which is not the highest either.

D – Here, the equilibrium is achieved where $AC = AR$, or in other words, when normal profits are made, and since the qty of “D” is sold, which is the highest, **option D becomes correct**, since the objective of “sales maximization” is achieved.

6-A:

- Allocative efficiency is NOT achieved, as firms produce at a level where price exceeds marginal cost. -> **Since they produce an output level which is less than the socially desirable level of output, there is an existence of deadweight loss of economic welfare.**
- Supernormal profits are NOT achieved, as firms earn normal profits due to free entry and exit in the long-run. This is shown through the fact that at the profit maximizing level of output, Average total cost is equal to Average revenue (price).



7-B:

The correct answer is B) imperfect competition and positive consumption externality.

Key take-away:- Positive consumption externality occurs when the consumption of a good or service benefits third parties, in this case, society as a whole.

The development and distribution of a vaccine by a few large pharmaceutical companies (imperfect competition) has a positive consumption externality, as it:

- Protects not only the individuals vaccinated but also those around them
- Reduces the risk of outbreaks and epidemics, benefiting society as a whole

Not only this, but the existence of **largest of firms**, indicates that market concentration is high, thus representing that there is **imperfect competition**.

A) Demerit good refers to a product with negative externalities, which doesn't apply here.

C) Perfect competition is unlikely in the pharmaceutical industry since there is **existence of largest of firms**, and negative consumption externality doesn't fit the scenario.

D) Public good is a product that is non-rivalrous and non-excludable, like national defense or public parks. Vaccines are a merit good and not public good.

8-C:

This option (**key C**) is not an example of the principal-agent problem because the tenants are consuming resources (electricity) that they are responsible for paying for (as stated in the initial description). The principal-agent problem typically involves a conflict of interest between the principal and agent, but in this case, the tenants are simply consuming resources they are responsible for. **This explains why key C will be correct.**

A) In this scenario, the tenants (agents) are responsible for managing the property (principal's asset) but their actions (forgetting to turn off the tap) lead to damage to the property (floor). This is an example of the principal-agent problem because the tenants' negligence affects the landlord's property, and the landlord may bear the cost of repairs. The tenants' actions are not aligned with the landlord's interests.

B) Here, the tenants (agents) are hosting a party, which may not be in the best interest of the landlord (principal). The guests' actions (damaging the interior) are a direct result of the tenants' decision to host a party, which may not align with the landlord's interests. The landlord may bear the cost of repairs, and the tenants' actions have negatively impacted the property.

D) In this scenario, the tenants (agents) are using the property (garden) in a way that damages it, which may not be in the best interest of the landlord (principal). The tenants' actions (parking their car) are not aligned with the landlord's interests, as they may lead to costly repairs of the garden.

9-C:

A positive net social benefit of \$50 million indicates that the **benefits of the project outweigh the costs, making it a worthwhile investment from a societal perspective.** However, the government may still choose not to proceed with the project if it has limited capital or budget constraints, and must prioritize other projects or expenditures. **This explains why key C is correct.**

Here's why the other options are incorrect:

A) The project's profitability for a private firm is not relevant to the government's decision, as the cost-benefit analysis suggests it has positive net social benefits. (this represents that $SB > SC$)

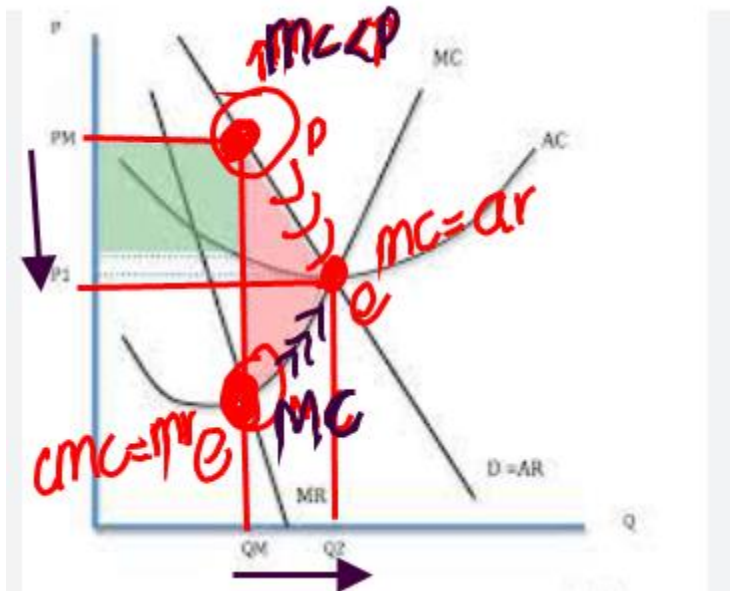
B) While negative externalities might be a concern, the positive net social benefit suggests that the benefits outweigh the costs, including any externalities.

D) The positive net social benefit suggests that the project's overall benefits exceed its costs

10-C:

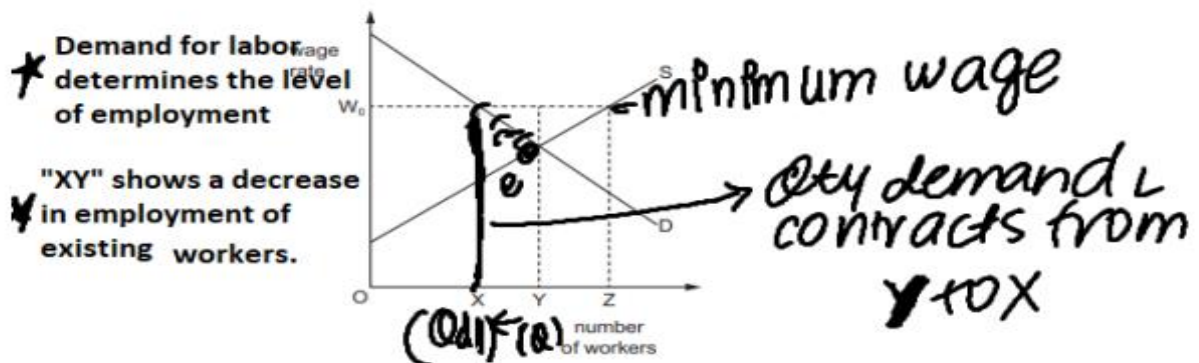
-> When the marginal cost is lower than the price, it means that the good is under-produced and undersupplied, leading to a shortage. To achieve allocative efficiency, the production and sale of the good should increase until the marginal cost equals the price

-> By increasing production and sale, the supply increases, and the price tends to fall until it equals the marginal cost, achieving allocative efficiency.



11-A:

The diagram shows an industry's demand for and supply of labour.



Initially, the labour market is in equilibrium. The government introduces a minimum wage of OW_0 .

What will be the effect on the level of employment in the industry?

- ☒ A a decrease equal to XY
- ☐ B a decrease equal to XZ
- ☐ C an increase equal to XY
- ☐ D an increase equal to YZ

12-D:

→ Differences in gradients represent changes in the price-elasticity of supply of labor.

A – This represents that market X has a higher demand for labor than market Y.

B – Since market X has a higher proportion of labor belonging to a trade union than Y, it represents that the workers working in the market X have a stronger negotiating power.

C – If market X has higher labor costs as a percentage of total costs than X, this represents that the workers belonging to the market X have an “elastic demand”. This becomes incorrect because we are interested in an option that explains the differences in the gradients of the labor supply curve.

D – If market X has more specific skills and training requirements than market Y, this represents that the supply of labor(working in market X) will be inelastic, because a rise in wages will result in a lesser proportionate increase in quantity supplied because the responsiveness of workers depends upon the time taken to receive training. This is why, key D, becomes correct.

13-D:

A – Although workers employed in occupation Y are given a “higher degree of job security”, it does not mean that **the workers attach a “greater importance to it”**. **Based on this, we cannot get to know about the individual preferences of workers working in the industry.**

B – Although, workers employed in occupation X have a working week of 48 hours, representing that they have a higher number of working hours as compared to occupation Y, this does not mean that they attach lesser importance to leisure activities. They may attach greater importance to leisure activities, but they may just not be able to get more leisure time.

C – We cannot deduce if there will be a greater competition for places on training courses to enter occupation X than occupation Y, since it depends upon the individual preferences of workers i.e. if they prefer financial advantages more than non-financial advantages.

D – This is correct to say because occupation Y offers more non-pecuniary or non-financial advantages as compared to occupation X. This is because, it offers a higher duration of annual leave in weeks, and also, the average length of working week is relatively lesser, and importantly, there is a higher degree of job-security available.

14-B:

- Direct taxes (like income tax) are progressive, meaning they take a greater share of income from the wealthy. If direct taxes fall, the wealthy will benefit more than the poor, increasing income inequality.

- Indirect taxes (like sales tax or VAT) are regressive, meaning they take a greater share of income from the poor. If indirect taxes rise, the poor will be disproportionately burdened, increasing income inequality.

This explains why key B will be correct.

15-D:.

To reduce the use of cars in cities, the government should encourage alternative modes of transportation, such as public transport. Subsidizing public transport makes it more affordable and attractive to people, thereby reducing the reliance on cars.

There is a positive relationship between the price of one good and the qty of another in case of substitutes. **When because of subsidies, the price of public transport decreases, the qty demanded of cars in cities will fall. This explains why key D will be correct.**

A) Decreasing parking charges would make parking easier and cheaper, encouraging more car use.

B) Lowering tax on petrol (gasoline) would make driving cheaper and more appealing, increasing car use.

C) Restricting cycling would discourage a sustainable and environmentally friendly mode of transportation, which is the opposite of what the government aims to achieve.

16-C:

Key take-away: Sales tax is not deducted from income, it's paid on consumption.

So, the correct calculation is:

$$\begin{aligned}\text{Disposable income} &= \text{Original income} - \text{Income tax} + \text{Cash benefits} \\ &= 100\% - 45\% + 5\% \\ &= 60\%\end{aligned}$$

17-B:

Option A:

- GDP decreases: Indicates a recession, leading to **lower tax revenues for the government.**
- Unemployment rises: More people are eligible for unemployment benefits, increasing **government spending.**
- Government spending decreases: A **reduction in discretionary government expenditure** reduces the impact of increased unemployment benefits, partially offsetting the revenue loss from lower GDP. **Therefore, this is why, key A becomes incorrect.**

Option B:

- GDP decreases: Lower tax revenues for the government.
- Unemployment rises: Increased government spending on unemployment benefits.
- Government spending remains unchanged: No reduction in discretionary expenditures to compensate for lower revenues. **This combination leads to a larger budget deficit.**

Option C:

- GDP increases: Higher tax revenues for the government.
- Unemployment falls: Reduced government spending on unemployment benefits.
- Government spending increases: Increased discretionary expenditures, potentially leading to a larger budget deficit.

However, the increase in GDP and resulting tax revenues would likely offset the increased spending, making it less likely to lead to a larger budget deficit. **This explains why key C will be incorrect.**

Option D:

- GDP increases: Higher tax revenues for the government.
- Unemployment falls: Reduced government spending on unemployment benefits.
- Government spending remains unchanged: No increase in discretionary expenditures to utilize the increased revenues.

This combination would lead to a smaller budget deficit or even a surplus, as revenues increase and discretionary government spending remains constant. **This explains why key D will be incorrect.**

18-C:

Option A:

- Central bank: Issues bank notes as part of its monetary policy mandate, regulating the money supply and maintaining financial stability.
- Commercial bank: They do not issue bank notes. Only central banks have the authority to print and distribute currency.

Option B:

- Central bank: Acts as lender of last resort, providing emergency loans to banks during financial crises.
- Commercial bank: Commercial banks are not lenders of last resort.

Option C:

- Central bank: Does not aim to maximize profit, since it is not a private owned bank.
- Commercial bank: Seeks to maximize profit through lending, as its primary objective being a **private sector firm** is to **maximize profits** to be able to generate returns for shareholders.

This explains why key C becomes correct.

Option D: This is incorrect because unlike a commercial bank, a central bank does receive deposits directly from the government.

19-B:

Option A: The challenges posed by urbanization include increased congestion, strain on city resources and infrastructure. Due to the resultant social issues, this will disrupt sustainable economic growth.

Option B: Technical training of the workforce are crucial for sustainable economic growth in a developed economy. This leads to improved productivity and efficiency, as workers acquire new skills and knowledge that enable them to produce more value-added goods and services. Technical training also increases innovation and competitiveness, as workers become more adept at problem-solving and adapting to new technologies. **Since the productive capacity of the economy increases which will benefit the future generations as well, it can be said that there will be “sustainable economic growth”.** This explains why key B will be correct.

Option C: A larger population puts pressure on resources, infrastructure, and the environment, potentially leading to decreased per capita income. **This can be seen in the graph below:**

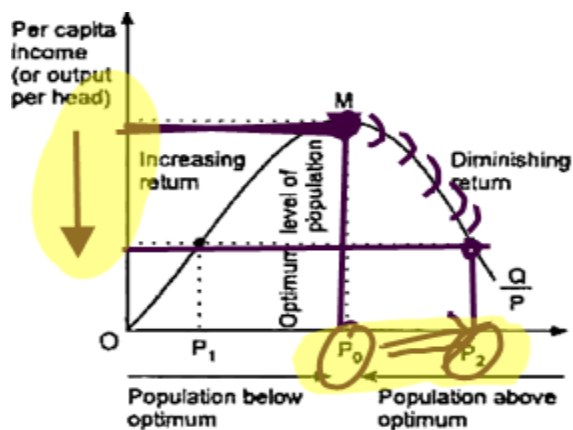


Fig. 4 : Optimum population

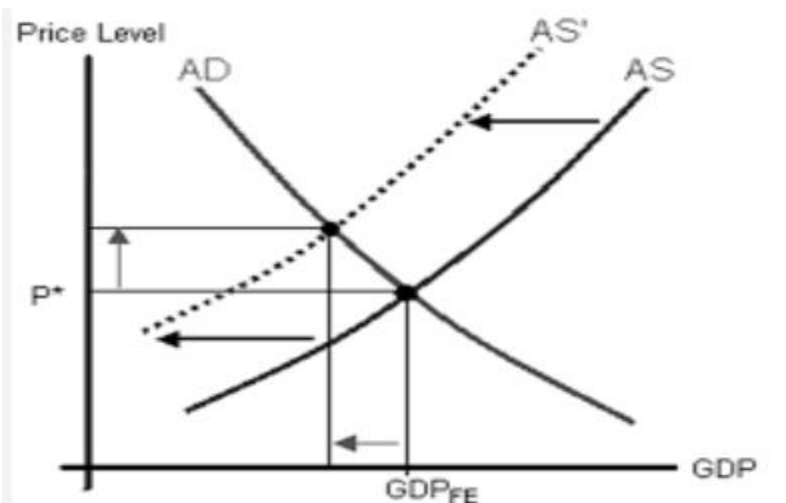
Additionally, a growing population increases the burden on public services and social security systems, potentially straining their capacity.

Option D: Rising consumer credit levels can boost consumption and economic activity in the short term, but it may also lead to debt accumulation and financial instability. As consumers take on more debt, they become more vulnerable to economic shocks and downturns, potentially leading to reduced spending and investment in the future. **This is how, sustainable economic growth will be compromised.**

20-B:

B) A significant increase in indirect taxes, such as value-added tax (VAT) or sales tax, can lead to higher production costs, as a result of which, Aggregate supply of the economy will fall. This will lead to stagflation being experienced because cost push inflation is accompanied with high levels of unemployment.

This can be seen in the diagram below:



A) An increase in retraining facilities can actually help make workers more flexible and versatile, because of which, when occupational mobility improves, they can easily and flexibly switch in between occupations especially when the structure of the economy is increasing. This therefore helps to reduce **structural unemployment**.

C) The imposition of selective import controls can protect domestic industries and jobs, potentially **reducing unemployment**.

D) An increase in the public sector deficit can lead to increased government spending which subsequently leads to an increase in Aggregate demand. This will bring about an increase in production and demand for labor, which will subsequently **reduce demand deficient unemployment**.

21-C:

The accelerator principle states that investment (I) is a function of the rate of change of income. This means that changes in **income** bring about **changes in induced investment**.

This is why, the value of the co-efficient of accelerator can be calculated using the formula given below:

Change in investment/ change in income.

This therefore, explains why key C will be correct.

A) When consumption is a function of a rate of change in income(change in consumption/change in income), this explains “**marginal propensity to consume**”

B) When income is a function of the rate of change of investment, this explains that changes in investment impact investment (change in income/ change in autonomous investment), which is why, it can be said that this explains the **investment multiplier**.

D) When investment depends upon the rate of interest, the concept of marginal efficiency of investment can be thought of.

22-B:

A) Monetary policy can influence employment and unemployment through interest rates and money supply management. By lowering interest rates, central banks can increase borrowing which can subsequently induce investment thereby increasing Aggregate demand. This way, they can reduce demand deficient unemployment.

C) Definition: Monetary policy involves the central bank making use of money supply, interest rates and exchange rate on behalf of the government to influence Aggregate demand.

They can intervene in the foreign exchange market to stabilize the exchange rate as part of monetary policy.

D) Monetary policy is often used to control inflation and maintain price stability. Central banks can use deflationary monetary policy i.e. raise interest rates to reduce borrowing and spending, thus reducing Aggregate demand and prices. By managing interest rates and money supply, central banks can influence the overall level of prices and achieve a steady price level, making it a key goal for monetary policy.

B) Monetary policy, which involves managing interest rates and money supply, is not an effective tool for achieving more equal income distribution. Income distribution is influenced by factors like taxation, labor market policies, and social welfare programs. **This explains why key B will be the correct answer.**

23-D)

- Full employment (yes): By increasing aggregate demand, expansionary fiscal policy can lead to increased employment and reduced **demand-deficient unemployment**, helping to achieve full employment.
- Stable price level (no): Expansionary fiscal policy can lead to increased Aggregate demand and when AD outpaces AS, there will be higher prices, potentially causing **demand-pull** inflation and deviating from the goal of price stability.
- Current account surplus (no): An expansionary fiscal policy can lead to increased incomes and imports will rise because of this, and a larger trade deficit will reduce the likelihood of a current account surplus.

24-A)

A) Devaluing the currency makes exports cheaper and more competitive in the global market. This increased competitiveness can lead to increased demand for exports, which can create employment opportunities in the export sector. **As a result, devaluation can have a positive effect on employment and economic growth, thus making key A correct.**

B) This option is a negative consequence of devaluation, as potential retaliatory measures can be taken by foreign countries. **If foreign importers depreciate their own currencies, it could offset the competitive advantage gained by country X's devaluation, potentially limiting the benefits of devaluation. (RETALIATION)**

C) Devaluation can make imports more expensive, leading to cost inflation if demand for imported raw materials is inelastic (i.e., not responsive to price changes). **This is because, when prices of imports rise, due to their inelastic nature, qty demanded will contract by a relatively smaller percentage, which will result in a rise in import expenditures. Subsequently, costs of production rise, and Aggregate supply falls, which translates into cost push inflation.**

D) Devaluation can lead to a loss of import duties as import demand contracts, which are a source of revenue for the government. This is again a disadvantage.

25-D:

-> Technological innovation can improve productivity, efficiency, and output. And since due to efficient production processes, the **long-run Aggregate supply or the productive potential of the economy increases, this represents that the living standards of the future generation will not be compromised.** This is why, there will be sustainable economic growth in case of option D.

This option involves the least cost for present and future generations, as it:

- Does not deplete natural resources (like mineral exploitation)-> **option A**
- Does not accumulate debt (like borrowing from abroad)-> **option B**
- Does not require significant sacrifices in present consumption (like high savings rates)-> **option C**

26-C:

Economic Growth: Refers to expansion of a country, measured by an increase in Real GDP. It mainly focuses on quantitative expansion. **That being said, real income per capita is included when measuring economic growth, which is why, it wouldn't be an additional component that may be included when measuring economic development. This is why, there will be a no under this category**

Economic Development:

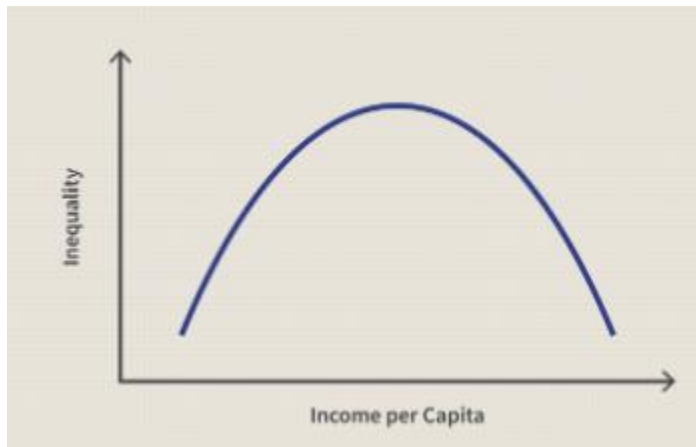
- Involves qualitative improvements in living standards, health, education, and income distribution
- Aims to enhance human development, well-being, and freedom

Therefore, to measure economic development, we need to see additional measures such as: how well the gdp is being distributed because that gives an insight about inequalities (changes in the distribution of gdp) and also the level of poverty because it will directly affect the well-being of people. Even if the economy is experiencing economic growth, there may still be absolute poverty faced by some groups which will hinder economic development. So to get a true idea of whether there is economic development or not, we need to account for the level of poverty as well.

This explains why key C will be correct.

27-A:

The **Kuznets curve** is a graphical representation of the relationship between economic growth and income inequality, and it is typically shaped like the curve shown in the diagram. The curve suggests that as economic growth increases, income inequality initially increases, but eventually decreases as the benefits of growth are shared more widely across the population.



The Lorenz curve (option c) is a graphical representation of income inequality, showing the proportion of total income earned by different segments of the population.

B) Laffer curve (option B): Illustrates the relationship between tax rates and government revenue, suggesting that beyond a certain point, higher tax rates can lead to lower revenue.

D) Philips Curve (option D) shows the tradeoff between unemployment and inflation, suggesting that lower unemployment can lead to higher inflation

28-D:

When evaluating the economic development of a country, we want to know how well it's doing in terms of prosperity, education, health, and overall well-being.

A) Adult literacy rate: Shows how many adults can read and write, indicating education levels.

B) Average life expectancy at birth: Shows how long people live on average, indicating healthcare quality.

C) Real GNP per capita: Shows the country's economic output per person, indicating income levels

D) Size of population: Just tells us how many people live in the country, not how well they're doing. A country with a large population might be poor, and a small country might be wealthy. This is why key D is correct.

29-C:

An expenditure-switching policy is a type of policy that aims to shift expenditure from imports to domestically produced goods and services, the effectiveness of which depends upon price elasticity of demand of domestic goods.

An increase in tariff rates is an example of such a policy, as it makes imports more expensive and encourages consumers to switch to domestic alternatives. **This is why, key C, becomes correct.**

Option A(an increase in income taxes) is a contractionary fiscal policy, and option B(an increase in interest rates) is a contractionary monetary policy. These policies are classed under the category of expenditure dampening policies, the target of which, is to reduce the consumption of imported goods.

Option D, as part of expansionary monetary policy, would increase incomes and encourage an increase in import spending.

30-C:

- Economic growth: Rise - The influx of migrant workers can increase the labor force in the high income country, as a result of which, there will be potential economic growth in the economy, since the productive capacity of the economy would now rise.
- Wage rises: Decrease - The increased labor supply can lead to a surplus of workers, which can drive down wages, especially in industries with a high concentration of migrant workers.

This explains why key C will be correct.

